

Instructions:

During a recent system compromise investigation, the Security Operations Center at Esquelle Eye Care discovered a critical oversight—the SIEM forwarding agent had never been installed on their corporate website's web server. This mistake left the web server's log data unmonitored.

A System Administrator managed to extract the Apache access log file from the server and has sent it to you for analysis. Your task is to review the challenge.log file to determine the extent of the compromise, its origin, the attack vector, and the vulnerability that led to the incident.

Challenge File:

04_SIEM/01_Log_Analysis/Challenges/challenge.log

Q1 How many total events were logged in the file?

343

Q2 What is the full timestamp of the last event in the file?

18/Jul/2024:15:59:25 -0400

Q3 What are the top 3 IP addresses found within the log file (in descending order)?

182.87.64.64, 53.64.228.139, 85.165.170.49

Q4 What is the count of the IP address with the highest connections?

23

Q5 How many different user agent strings are found in the log file?

3

Q6 Which user agent string sticks out as suspicious?

Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/126.0.0.0 Safari/537.36

Q7 What is the line number of the log event that returned an HTTP status code of 404?

76

Q8 What web-based attack can you identify based on the URL query parameters?

SQL Injection

Q9 What is the name of the vulnerable URL parameter?

q

Q10 The attacker attempted a number of different SQL injection attacks. Based on other indicators in the event logs, what was the full timestamp of the potentially successful attempt?

18/Jul/2024:15:58:52 -0400

Challenge URL

<https://challenges.malwarecube.com/#/c/d63f1671-a312-4677-8684-3cf446870536>